Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7 HT	(a)	(i)		**Indicative content** Early on He and H_2 escaped from atmosphere. Water vapour present in early atmosphere (from volcanoes) condensed to form oceans. (As the earth cooled) the % of carbon dioxide has decreased to a fraction of 1%. This is due to: photosynthesis in green plants, using up carbon dioxide and releasing oxygen; More CO_2 locked up by animals evolving shells; subsequent formation of limestone; Carbon lock up in fossil fuels from the remains of simple marine organisms. Ammonia decomposed by a reaction with oxygen forming nitrogen the most abundant gas in the atmosphere. Reference to time 4.5-4.0 billion years ago and present day. **5– 6 marks** Candidates can give a comprehensive description of the changing composition of the atmosphere. They recognise the major changes in all gases. They give a comprehensive explanation of why these changes have occurred from early on to present day. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3 – 4 marks** Candidates can quantitatively (time or %) describe some changes in the atmosphere. OR Candidates can give some explanation or reasons for the changes in water vapour, carbon dioxide or oxygen. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology*		2	4	6		

				and some accurate spelling, punctuation and grammar. **1-2 marks** Candidates can describe some of the changes that have occurred in the atmosphere. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** No attempt made or no response worthy of credit.						
	(b)	(i)		Compounds found on early Earth contained Carbon, Hydrogen, Nitrogen and Oxygen (1) Required to form (named) biological molecules (1)			2	2		
		(ii)		Earth was hot (1) increased electrical activity (lightning) on Earth (1)			2	2		
				Question 7 total	**0**	**2**	**8**	**10**	**0**	**0**